

DK

Retail Strategy & Analytics Chip Category Review

Quantium Data Analytics Job Simulation

Customer Segmentation | Experimentation | Uplift Testing

264,834

Transactions analysed

72,637

Unique customers

12

Months of data

3

Trial stores assessed

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R | Python | SQL | Advanced Excel

github.com/kamdardhruv8-droid/quantium-virtual-internship

Executive Summary

Analysed 264,834 chip transactions across 72,637 customers (Jul 2018 to Jun 2019) to identify high-value customer segments and evaluate a store layout trial. The analysis combined data preparation, customer segmentation, statistical testing, and experimentation to deliver actionable recommendations for the Category Manager.

Customer Segments

Three segments drive 26% of total chip revenue: Budget Older Families (\$160K), Mainstream Young Singles/Couples (\$149K), and Mainstream Retirees (\$147K). Mainstream Young Singles/Couples represent the highest-value opportunity. They pay the most per unit (\$4.05 vs \$3.65 average), over-index on premium brands (Tyrrells +23%, Kettle +20%, Pringles +19%), and prefer larger pack sizes (270g+). This spending pattern is statistically significant.

Trial Store Performance

All three trial stores (77, 86, 88) showed statistically significant sales uplift in at least 2 of 3 trial months (Feb to Apr 2019), driven primarily by increased customer traffic. Stores 86 and 88 had significantly more customers in all three trial months. The trial layout reversed declining sales trends and attracted new shoppers to the chip category.

Recommendation

Roll out the trial layout to high-traffic stores. Target Mainstream Young Singles/Couples with premium brand promotions (Tyrrells, Kettle, Pringles) in 270g+ packs positioned in high-visibility locations.

Data Preparation and Quality

The transaction dataset required significant cleaning before analysis. Product names contained inconsistent brand abbreviations (WW for Woolworths, RRD for Red Rock Deli, Smith vs Smiths), embedded weight values in mixed formats (175g, 330G), and non-chip products (salsa dips) that needed removal.

Cleaning Steps

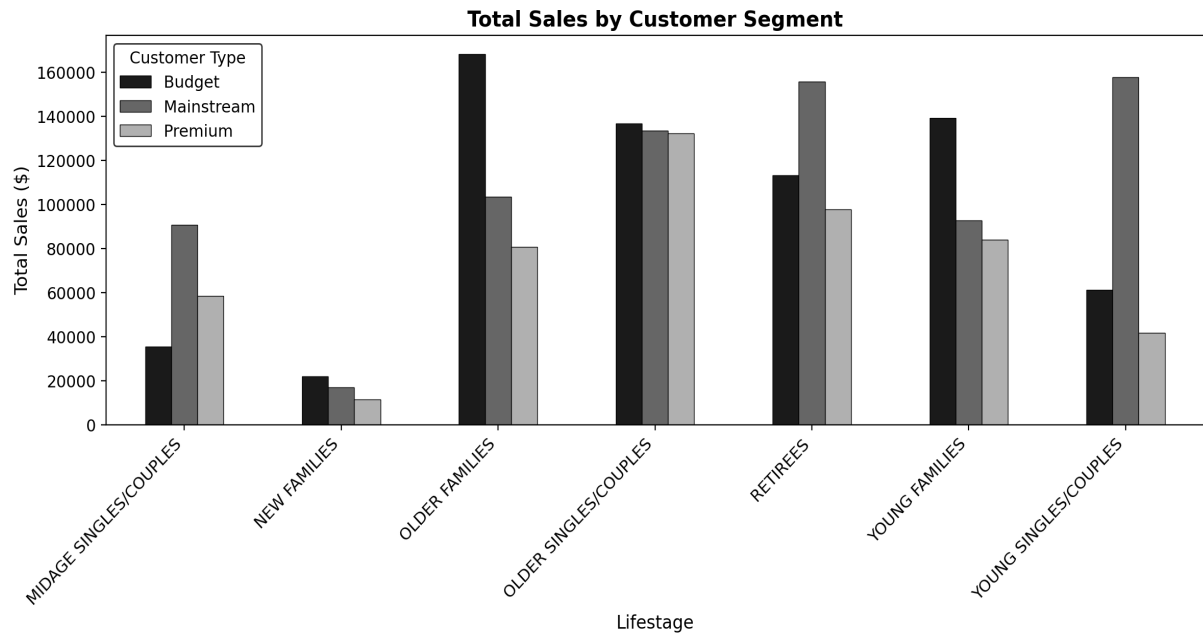
Extracted pack sizes from product names using regex. Standardised 28 brand abbreviations across 114 unique products. Exported unique products to Excel for manual validation, then re-mapped to the full dataset with zero unmatched entries. Removed 5 salsa dip products (Old El Paso, Woolworths) that are not chips. Identified and removed 1 commercial buyer (customer 226000: 200 units per transaction, 2 transactions only). Verified all numeric columns, fixed date format from Excel serial numbers, and confirmed zero null values after merging with customer data.

Step	Action	Result
Weight extraction	Regex on PROD_NAME	21 unique pack sizes (70g to 380g)
Brand standardisation	gsub on 28 abbreviations	20 clean brand names
Non-chip removal	Keyword filter (SalsaDip, MildSalsa, MediumSalsa)	5 products removed, 4 salsa-flavour chips retained
Outlier removal	Customer 226000 flagged	2 rows removed (400 units total)
Merge validation	Left join on LYLTY_CARD_NBR	252,589 rows, zero NAs

Category Overview: Who Buys Chips

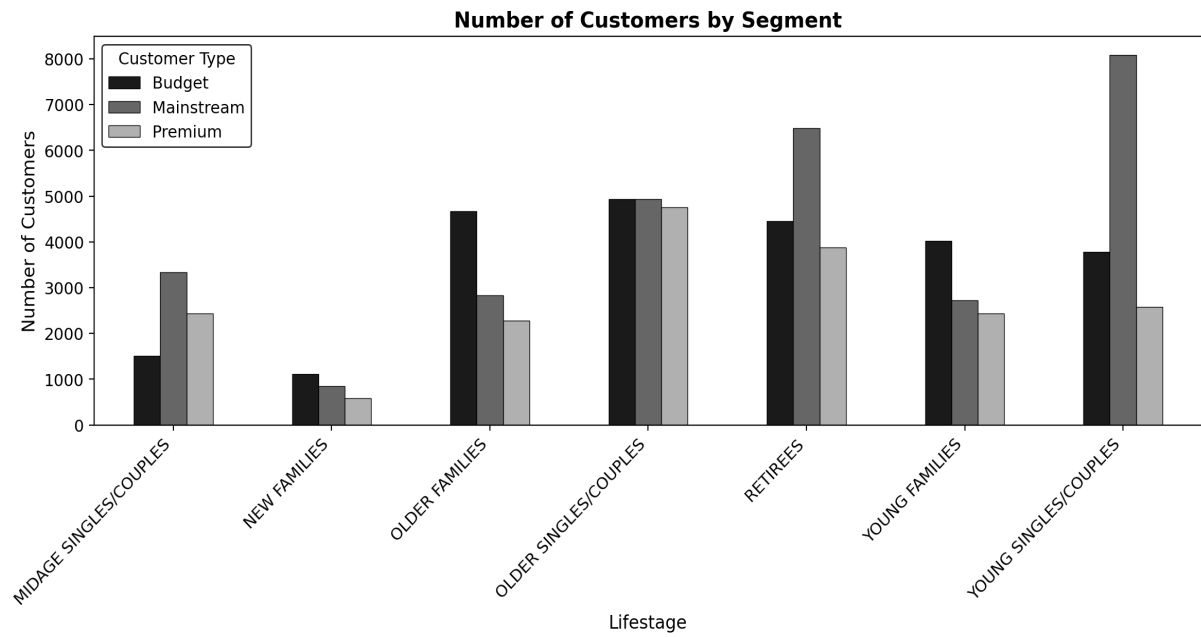
Customer segments were analysed across 7 lifestages and 3 purchasing tiers (Budget, Mainstream, Premium), producing 21 unique segments. Total sales, customer counts, average units per customer, and average price per unit were calculated for each.

Total Sales by Segment



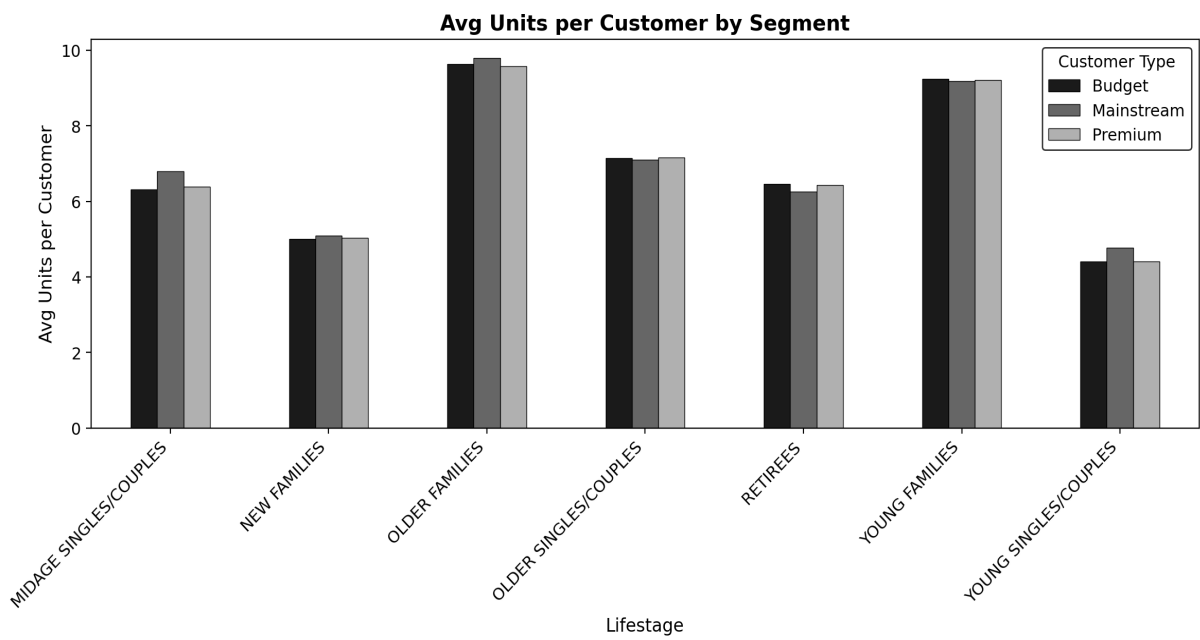
Budget Older Families lead at \$160K, followed by Mainstream Young Singles/Couples (\$149K) and Mainstream Retirees (\$147K).

Number of Customers by Segment



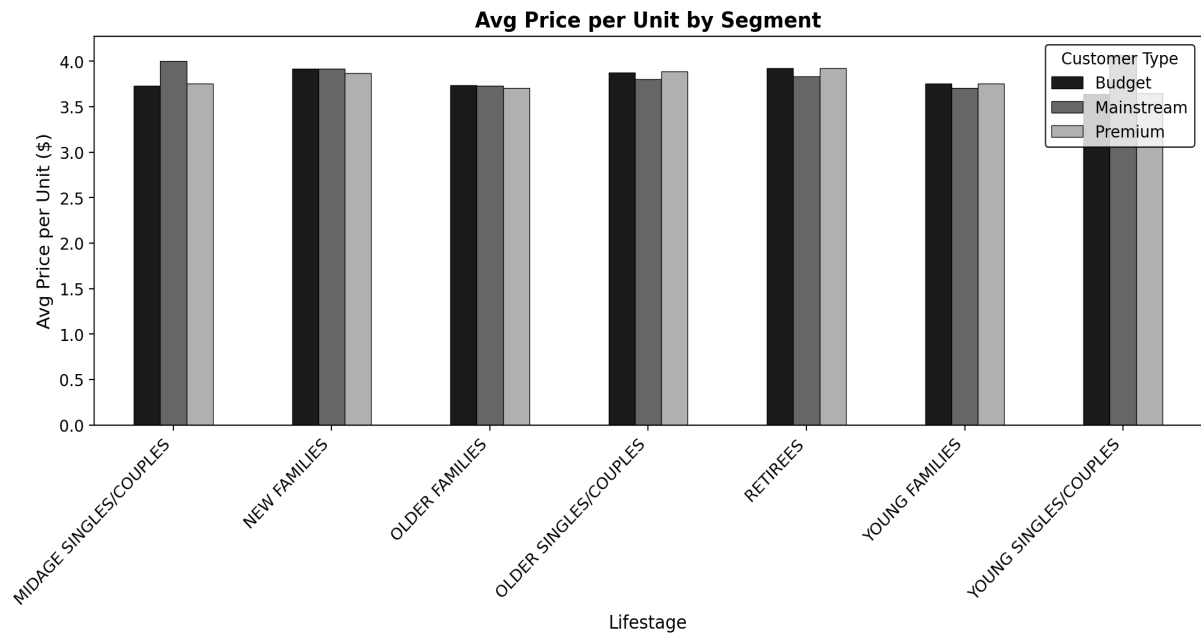
Mainstream Young Singles/Couples are the largest group at 7,943 customers, explaining their high total sales. Budget Older Families achieve the highest revenue with fewer customers (4,635), indicating higher spend per customer.

Average Units per Customer



*Older Families and Young Families buy 9 to 10 units per customer, driven by larger household sizes.
Young Singles/Couples buy 4 to 5 units but pay more per unit.*

Average Price per Unit

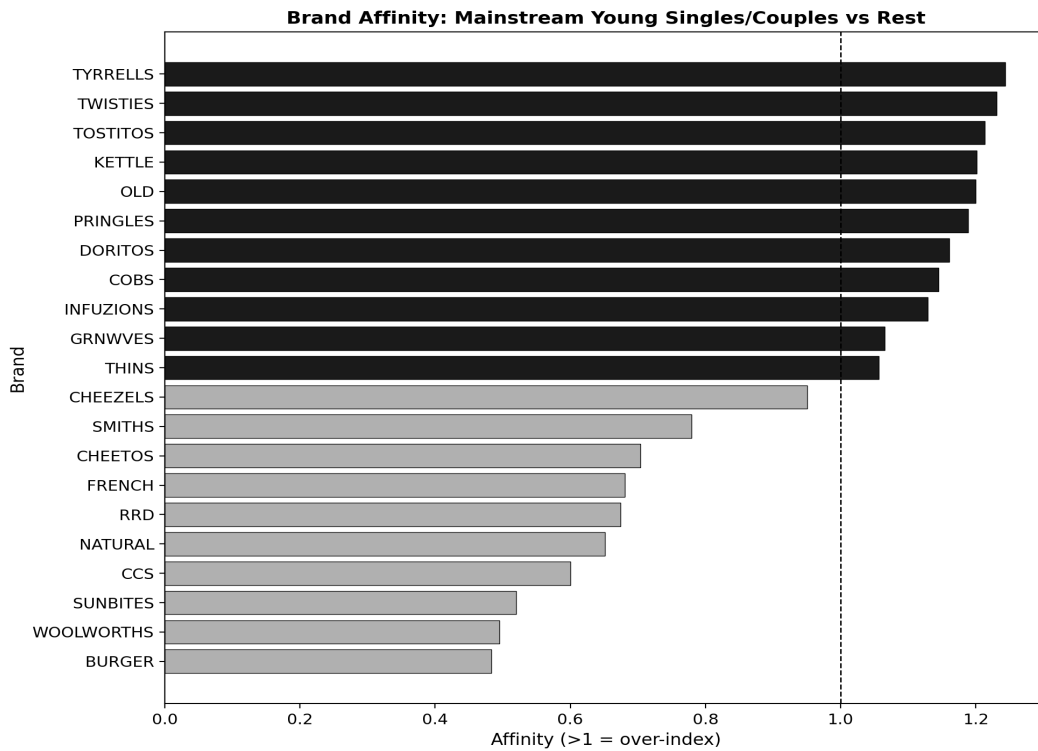


Mainstream Young Singles/Couples pay \$4.05 per unit vs \$3.65 for Budget and Premium counterparts. This difference is statistically significant (Welch t-test, $t = 38.6$, $df = 56,388$). This suggests impulse buying behaviour rather than price-driven purchasing.

Deep Dive: Mainstream Young Singles/Couples

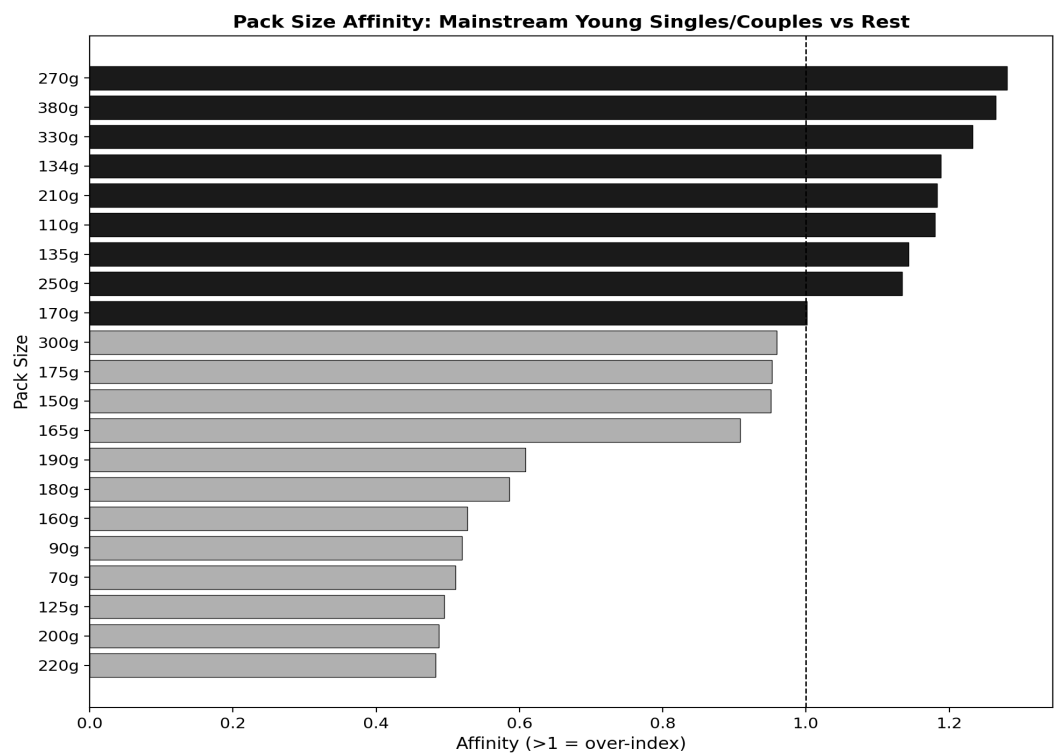
Affinity analysis compares this segment's purchasing patterns against the rest of the population. An affinity score above 1.0 means the segment buys proportionally more of that brand or pack size.

Brand Affinity



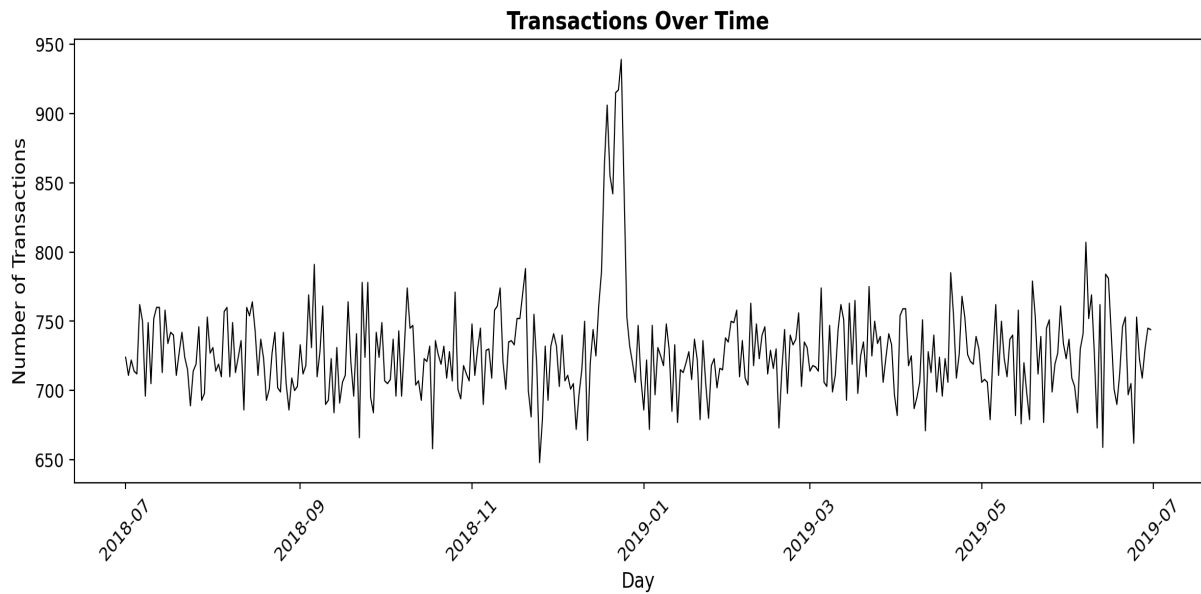
Over-indexed brands: Tyrrells (+23%), Twisties (+23%), Tostitos (+21%), Kettle (+20%), Pringles (+19%). Under-indexed: Woolworths (-51%), BurgerRings (-52%), CCs (-40%). This segment gravitates toward brands positioned as premium or trendy.

Pack Size Affinity



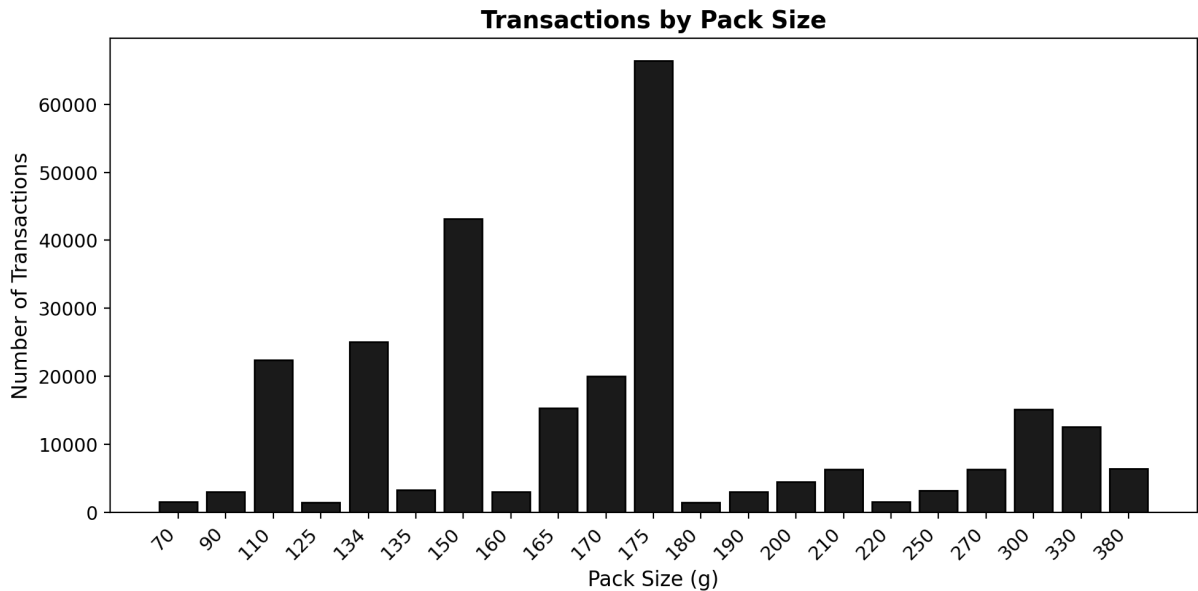
Strongly prefers larger packs: 270g (+28%), 380g (+27%), 330g (+23%). Avoids smaller economy sizes: 70g (-49%), 90g (-48%), 125g (-50%). Combined with brand preference, this segment buys larger packs of premium brands.

Seasonal Transaction Patterns



Transactions are stable at approximately 700 to 750 per day throughout the year. A clear spike occurs in mid-December (pre-Christmas shopping), peaking around December 21 to 22. Zero transactions on December 25 (Christmas Day store closures). Only 364 of 365 days have recorded transactions.

Pack Size Distribution



175g dominates with over 66,000 transactions. 150g and 134g follow. The bulk of volume sits in the 110g to 175g range, but larger sizes (250g+) carry higher per-transaction value.

Trial Store Performance Assessment

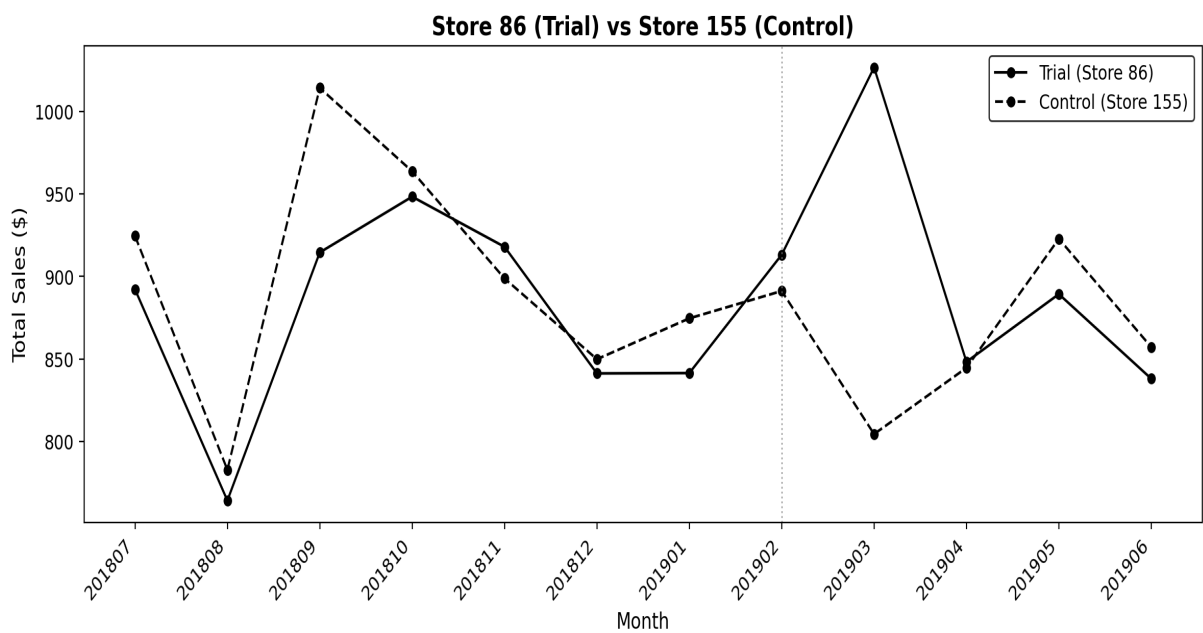
Stores 77, 86, and 88 underwent a layout trial. Control stores were selected using pre-trial data (Jul 2018 to Jan 2019) through a two-stage process: Pearson correlation screening (threshold > 0.5) followed by normalised magnitude distance scoring on monthly sales and customer counts. The formula used for distance: $1 - (\text{observed distance} - \text{minimum distance}) / (\text{maximum distance} - \text{minimum distance})$. Final score: $0.5 * \text{correlation} + 0.5 * \text{normalised magnitude}$, averaged across sales and customers.

Trial Store	Control Store	Selection Score	Method
77	233	0.95	Pre-trial correlation + magnitude
86	155	0.93	Pre-trial correlation + magnitude
88	178	0.93	Pre-trial correlation + magnitude

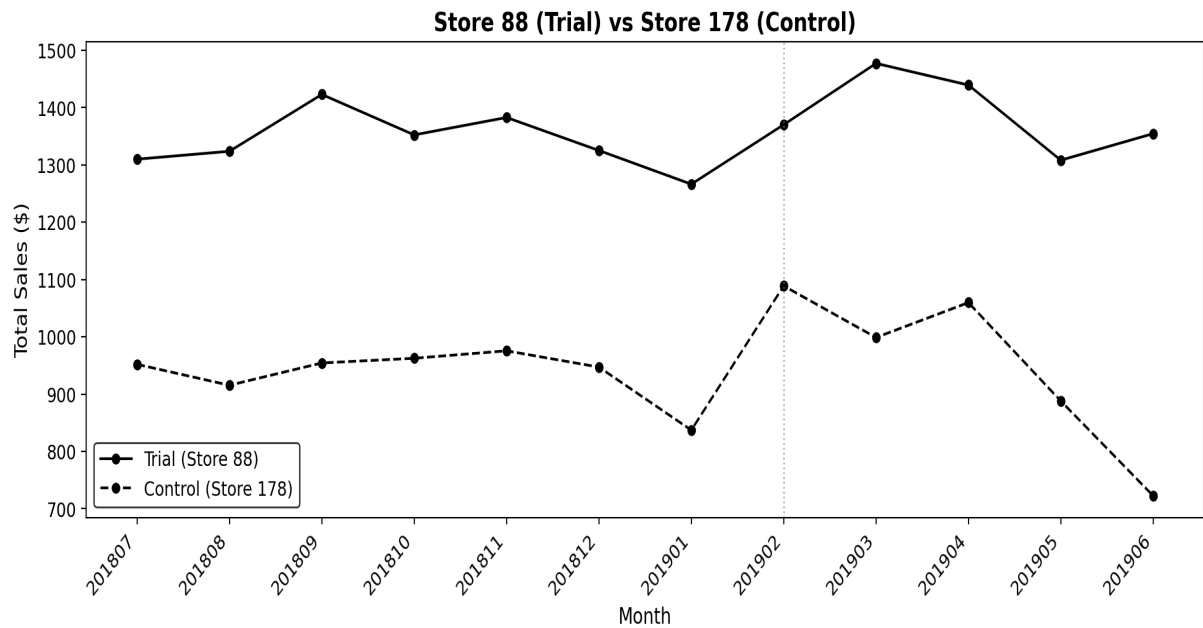
Store 77 (Trial) vs Store 233 (Control)



Store 86 (Trial) vs Store 155 (Control)



Store 88 (Trial) vs Store 178 (Control)



Statistical Significance

Each trial store was assessed against its scaled control using percentage difference and t-values against the 95th percentile of the t-distribution (7 degrees of freedom, threshold = 1.89).

	Store 77 vs 233	Store 86 vs 155	Store 88 vs 178
Total Sales	2/3 months significant	2/3 months significant	2/3 months significant
Customer Count	2/3 months significant	3/3 months significant	3/3 months significant
Primary Driver	Moderate uplift	More customers	More customers

Self-comparison confirmed the trial reversed declining trends: Store 77 was declining at -\$4.01/month before the trial. Store 86 was declining at -\$6.78/month. Store 88 had modest growth of \$5.57/month that accelerated significantly during the trial.

Recommendations

Roll out the trial layout

Deploy to stores with similar profiles to stores 86 and 88, where the effect was strongest. Prioritise high-traffic locations where increased customer flow translates to the largest absolute sales uplift.

Target Mainstream Young Singles/Couples

Position Tyrrells, Kettle, and Pringles in 270g+ packs in high-visibility, impulse-purchase locations. This segment pays 11% more per unit and is the largest customer group.

Investigate Store 86 pricing

Customer counts were significantly higher in all 3 trial months, but sales significance dropped in April. Check whether promotional pricing offset the customer uplift.

Seasonal promotions

Plan chip promotions for early to mid December to capitalise on the pre-Christmas transaction spike.

Next phase support

Quantum can identify optimal store clusters for rollout, measure post-rollout impact, and refine targeting by tracking segment-level response to the new layout.

Tools and Methodology

Area	Tools / Techniques
Data preparation	R (dplyr, stringr, readxl), Excel
Visualisation	R (ggplot2), Python (matplotlib)
Statistical testing	Welch two-sample t-test, confidence intervals, t-distribution
Segmentation	LIFESTAGE x PREMIUM_CUSTOMER cross-tabulation
Control store selection	Pearson correlation + normalised magnitude distance
Affinity analysis	Segment share / population share ratio
Reporting	PowerPoint (Pyramid Principle framework), PDF